

The purpose of this note is twofold. First, we explain the dx notation that we use in class for continuous random variables and discuss how useful it is when computing the distribution of a function of a random variable. In the second part, we will describe a general “change of variables” method of computing the distribution of a function of a random variable (or a random vector).

In sections 1 and 2, we introduce the dx notation for continuous random variables and we compare its usage against what happens for discrete random variables. In section 3, some methods for computing densities of random variables are discussed. The dx notation is used to define and compute conditional probabilities and expectations in Section 4.

1 Real-valued random variables

In this course, we use two types of random variables: discrete and continuous.

1.1 Discrete

Recall that a random variable X is called discrete if it takes values in a countable set $\mathcal{D} = \{a_1, a_2, \dots\}$. The probabilities that X takes each of these values $\mathbf{P}\{X = a_1\}$, $\mathbf{P}\{X = a_2\}, \dots$ sum to 1:

$$\sum_{i=1}^{\infty} \mathbf{P}\{X = a_i\} = 1. \quad (1)$$

The collection of the probabilities $\mathbf{P}\{X = a_1\}, \mathbf{P}\{X = a_2\}, \dots$ is called the distribution of X . Using the distribution of X , we can calculate the probability $\mathbf{P}\{X \in A\}$ for an arbitrary set of values $A \subset \mathcal{D}$:

$$\mathbf{P}\{X \in A\} = \sum_{a \in A} \mathbf{P}\{X = a\}.$$

Moreover, we can calculate the expected value of a (numerical) function of X . Given a function $\varphi: \mathcal{D} \rightarrow \mathbb{R}$, we define

$$\mathbf{E}(\varphi(X)) = \sum_{i=1}^{\infty} \varphi(a_i) \mathbf{P}\{X = a_i\}. \quad (2)$$

In particular, if the values of X are real numbers, that is, $\mathcal{D} \subset \mathbb{R}$, then the expectation of X is given by

$$\mathbf{E}(X) = \sum_{i=1}^{\infty} a_i \mathbf{P}\{X = a_i\}.$$

1.2 Continuous

Let us now assume that X is a random variable that takes values in an uncountable set $\mathcal{D} \subset \mathbb{R}$. (For instance, $\mathcal{D} = [0, +\infty)$ if X is an exponential random variable and $\mathcal{D} = \mathbb{R}$ if X is a Gaussian random variable.) In this course, we deal with *continuous* random variables which have zero probability of taking a fixed value: $\mathbf{P}\{X = a\} = 0$ for every $a \in \mathcal{D}$. Thus, the probabilities that X takes each fixed value do not help in computing/defining the probability $\mathbf{P}\{X \in A\}$ for an arbitrary set $A \subset \mathcal{D}$ or the expectation $\mathbf{E}(\varphi(X))$ for an arbitrary function $\varphi: \mathcal{D} \rightarrow \mathbb{R}$. Instead, we introduce the notion of *density* which is also the defining property of a *continuous* random variable.

Definition 1. A random variable X is called continuous if there exists a function $f: \mathbb{R} \rightarrow \mathbb{R}_+$ with $\int_{-\infty}^{\infty} f(x)dx = 1$ such that

$$\mathbf{P}\{X \in [a, b]\} = \int_a^b f(x)dx \quad (3)$$

for every $a, b \in \mathbb{R}$ with $a \leq b$. In this case, f is called the density of X .

If we take $a = x, b = x + \Delta x$ where $\Delta x > 0$ is a very small number, then on $[a, b] = [x, x + \Delta x]$, the values of f are approximately constant and equal to $f(x)$. Therefore,

$$\mathbf{P}\{X \in [x, x + \Delta x]\} = \int_x^{x+\Delta x} f(y)dy \approx f(x) \int_x^{x+\Delta x} dy = f(x)\Delta x. \quad (4)$$

This explains the name “density”: while $f(x)$ is not a probability itself, we can interpret $f(x)$ as the “probability per unit length around x .”

The dx notation provides a convenient way of writing densities: if X is a continuous random variable with density f , then we write

$$\mathbf{P}\{X \in [x, x + dx]\} = f(x)dx, \quad x \in \mathbb{R}. \quad (5)$$

Note that (5) is very similar to the approximation (4), which makes the notation intuitive. In (5), $[x, x + dx]$ can be thought of as an *infinitesimal* interval around x that is of length dx . To avoid the lengthy notation $[x, x + dx]$, with a slight abuse of notation, we will denote $[x, x + dx]$ by dx :

$$\mathbf{P}\{X \in dx\} = f(x)dx, \quad x \in \mathbb{R}.$$

While dx denotes the infinitesimal interval on the left hand side, the dx on the right hand side is the length of that interval.

Now that we have the dx notation, we can write

$$\mathbf{P}\{X \in [a, b]\} = \int_a^b \mathbf{P}\{X \in dx\}.$$

If we take $a = -\infty, b = \infty$, then this gives

$$\int_{-\infty}^{\infty} \mathbf{P}\{X \in dx\} = \mathbf{P}\{X \in [-\infty, \infty]\} = 1,$$

which looks very similar to the discrete version (1). We also define the expectation of $\varphi(X)$ as

$$\mathbf{E}(\varphi(X)) = \int_{-\infty}^{\infty} x \mathbf{P}\{X \in dx\} = \int_{-\infty}^{\infty} x f(x) dx,$$

again, this looks very close to its discrete analog (2).

Note that the dx notation is only a convenient way of writing the density but not a substitute for it. We still need the density function to do actual computations.

1.3 Optional: neither discrete nor continuous

You may wonder if there exists a random variable that has uncountably many possible values but still has $\mathbf{P}\{X = a\} > 0$ for at least one possible value a (hence it does not have a density). Indeed, it is not very difficult to define such a random variable using the tools of this course. Although we shall not encounter such random variables later, let us give one example for the sake of completeness.

Let Z be a Bernoulli variable with success probability $\frac{1}{2}$. Let X be another random variable. Suppose that X is conditionally Expon(1) given that $Z = 1$, and it is conditionally deterministic and equals zero given that $Z = 0$. In other words,

$$\begin{aligned} \mathbf{P}\{X \in dx | Z = 1\} &= e^{-x} dx, \quad x \geq 0, \\ \mathbf{P}\{X = 0 | Z = 0\} &= 1. \end{aligned}$$

Now what is $\mathbf{P}\{X = 0\}$? This is easy to compute by conditioning:

$$\begin{aligned} \mathbf{P}\{X = 0\} &= \mathbf{P}\{X = 0 | Z = 1\} \mathbf{P}\{Z = 1\} + \mathbf{P}\{X = 0 | Z = 0\} \mathbf{P}\{Z = 0\} \\ &= \int_0^0 e^{-x} dx \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} \\ &= 0 + \frac{1}{2} = \frac{1}{2} > 0. \end{aligned}$$

Since $\mathbf{P}\{X = 0\} > 0$, X cannot have a density, so it is not a continuous random variable according to Definition 1. On the other hand, X can take any value in the uncountable set $[0, \infty]$, so it is not a discrete random variable. Therefore, X is neither discrete nor continuous! However, it is conditionally continuous given $Z = 1$ and conditionally discrete (and deterministic) given $Z = 0$.

2 Random vectors: joint distributions

Let X, Y be random variables with values in $\mathbb{R}$. We call the pair (X, Y) a random vector. In this note, we only talk about two-dimensional random vectors. The general-

ization to higher dimensions is straightforward.

2.1 Discrete

Assume that both X, Y are discrete: X takes values in $\mathcal{D}_X = \{a_1, a_2, \dots\}$ and Y takes values in $\mathcal{D}_Y = \{b_1, b_2, \dots\}$. Then

$$\sum_{i,j \geq 1}^{\infty} \mathbf{P}\{X = a_i, Y = b_j\} = 1.$$

The collection $\{\mathbf{P}\{X = a_i, Y = b_j\}\}_{i,j \geq 1}$ is called the *joint distribution* of X, Y . Using the joint distribution, we can calculate

$$\mathbf{P}\{X \in A, Y \in B\} = \sum_{a \in A, b \in B} \mathbf{P}\{X = a, Y = b\}$$

for arbitrary subsets $A \subset \mathcal{D}_X$ and $B \subset \mathcal{D}_Y$. In particular, we can calculate the (marginal) distribution of X by

$$\mathbf{P}\{X = a_i\} = \sum_{j=1}^{\infty} \mathbf{P}\{X = a_i, Y = b_j\}, \quad i \geq 1,$$

and similarly the distribution of Y . Besides, for a bivariate function $\varphi: \mathcal{D}_X \times \mathcal{D}_Y \rightarrow \mathbb{R}$, we can calculate the expectation of the random variable $\varphi(X, Y)$ by

$$\mathbf{E}(\varphi(X, Y)) = \sum_{i,j \geq 1} \varphi(a_i, b_j) \mathbf{P}\{X = a_i, Y = b_j\}.$$

2.2 Continuous

As in one-dimensional case, let us carefully define what we mean by a continuous random vector. This will bring us to the notion of *joint density*.

Definition 2. A random vector (X, Y) is called *jointly continuous* if there exists a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_+$ with $\int_{x,y \in \mathbb{R}} f(x, y) dx dy = 1$ such that

$$\mathbf{P}\{X \in [a, b], Y \in [c, d]\} = \int_c^d \int_a^b f(x, y) dx dy, \quad (6)$$

for every $a, b, c, d \in \mathbb{R}$ with $a \leq b, c \leq d$. In this case, f is called the *joint density* of X, Y .

The two dimensional version of the dx notation is as follows: if (X, Y) is a jointly continuous random vector with density f , then we write

$$\mathbf{P}\{X \in dx, Y \in dy\} = f(x, y) dx dy, \quad x \in \mathbb{R}, y \in \mathbb{R}.$$

The infinitesimal interval/length interpretations of dx, dy are the same as in the one-dimensional case.

With the dx notation, we can rewrite (6) as

$$\mathbf{P}\{X \in [a, b], Y \in [c, d]\} = \int_{x \in [a, b]} \int_{y \in [c, d]} \mathbf{P}\{X \in dx, Y \in dy\},$$

where we write the name of the variable for each integral to avoid confusion. Note that if we take $c = -\infty, d = \infty$, we get

$$\mathbf{P}\{X \in [a, b]\} = \int_{x \in [a, b]} \int_{y \in [-\infty, \infty]} \mathbf{P}\{X \in dx, Y \in dy\}.$$

On the other hand, we have

$$\mathbf{P}\{X \in [a, b]\} = \int_{x \in [a, b]} \mathbf{P}\{X \in dx\}.$$

Therefore, the (marginal) density of X is given by

$$\mathbf{P}\{X \in dx\} = \int_{y \in \mathbb{R}} \mathbf{P}\{X \in dx, Y \in dy\}.$$

With the conventional density notation, we have

$$\mathbf{P}\{X \in dx\} = \int_{y \in \mathbb{R}} f(x, y) dx dy = \left(\int_{y \in \mathbb{R}} f(x, y) dy \right) dx,$$

that is, the density of X is the function $x \mapsto \int_{y \in \mathbb{R}} f(x, y) dy$. In particular, we have shown that if (X, Y) is a jointly continuous random vector, then X and Y are continuous random variables.

Finally, we calculate the expectation of $\varphi(X, Y)$ by

$$\mathbf{E}(\varphi(X, Y)) = \int_{x, y \in \mathbb{R}} \varphi(x, y) \mathbf{P}\{X \in dx, Y \in dy\} = \int_{x, y \in \mathbb{R}} \varphi(x, y) f(x, y) dx dy.$$

2.3 Mixed types

Here is one case which is not possible to deal with using the conventional density notation. We assume that X is a discrete random variable and Y is a continuous random variable. Suppose that they are independent. How can we write the joint distribution of X, Y . Clearly, (X, Y) is not discrete as the second component Y can take uncountably many values. On the other hand, (X, Y) is not jointly continuous as otherwise X would be continuous as we argued in the previous section. Using the dx notation, we can write the *joint distribution* of X, Y as

$$\mathbf{P}\{X = a_i, Y \in dy\} = \mathbf{P}\{X = i\} \mathbf{P}\{Y \in dy\}.$$

One might argue that since X, Y are independent, there is no need to use the complicated notation $\mathbf{P}\{X = a_i, Y \in dy\}$ and we can use the product form $\mathbf{P}\{X = i\}\mathbf{P}\{Y \in dy\}$. Let us consider another example of a “mixed” random vector (X, Y) where X, Y are not independent. Suppose that $Y \sim \text{Expon}(1)$ and given that $Y = y$, the conditional distribution of X is $\text{Pois}(y)$. Hence,

$$\begin{aligned}\mathbf{P}\{Y \in dy\} &= e^{-y}dy, & y \geq 0, \\ \mathbf{P}\{X = k|Y = y\} &= \frac{e^{-y}y^k}{k!}, & k \in \{0, 1, 2, \dots\}.\end{aligned}$$

Clearly, X is discrete and Y is continuous. Their joint distribution is given by

$$\mathbf{P}\{X = k, Y \in dy\} = \mathbf{P}\{X = k|Y = y\}\mathbf{P}\{Y \in dy\} = \frac{e^{-2y}y^k}{k!}dy.$$

In general, if the joint distribution of X, Y has the mixed form $\mathbf{P}\{X = a_i, Y \in dy\}$, the marginal distributions of X, Y are given by

$$\begin{aligned}\mathbf{P}\{X = a_i\} &= \int_{y \in \mathbb{R}} \mathbf{P}\{X = a_i, Y \in dy\}, \\ \mathbf{P}\{Y \in dy\} &= \sum_{i=1}^{\infty} \mathbf{P}\{X = a_i, Y \in dy\}.\end{aligned}$$

3 Distribution of a function of a random variable

Let X be a random variable whose distribution is known. Let $Y = h(X)$ be another random variable that is a function of X , e.g., $Y = X + 5$, $Y = X^2$, $Y = \sin(X)$. In this section, we are interested in finding the distribution of Y .

3.1 Some heuristics

Suppose that h is a smooth one-to-one function whose inverse is h^{-1} . If X is discrete, it is very easy to compute the distribution of Y (in terms of the distribution of X):

$$\mathbf{P}\{Y = b\} = \mathbf{P}\{h(X) = b\} = \mathbf{P}\{X = h^{-1}(b)\}.$$

If X is continuous, it is not clear what the analog of the above computation should be! To make the point, let us assume that X is a positive random variable with density f and define $Y = h(X) = X^2$. Let us try to mimic the above computation:

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\{X^2 \in dy\} \stackrel{?}{=} \mathbf{P}\{X \in \sqrt{dy}\} \stackrel{?}{=} f(\sqrt{y})\sqrt{dy}.$$

However, $\sqrt{dy}$ does not make any sense as the square root of an infinitesimal interval/length!

In general, we cannot manipulate the dx notation by freely taking square roots, cubes, sines, etc. The distribution of $Y = h(X)$ can be computed using the dx notation *only* under two special cases: shifting, scaling by a positive number.

3.1.1 Shifting

Suppose that X is a continuous random variable with density f and define $Y = X + a$ for some $a \in \mathbb{R}$. To compute the density of Y , that is,

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\{X + a \in dy\},$$

we recall that dy is an infinitesimal interval around y of length dy . If $X + a$ takes values in an interval around y of length dy , then X must take values in an interval around $y - a$ of length dy :

$$\begin{aligned} \mathbf{P}\{Y \in dy\} &= \mathbf{P}\{X + a \in [y, y + dy]\} \\ &= \mathbf{P}\{y \leq X + a \leq y + dy\} \\ &= \mathbf{P}\{y - a \leq X \leq (y - a) + dy\} \\ &= \mathbf{P}\{X \in [y - a, (y - a) + dy]\}. \end{aligned}$$

We will denote the interval $[y - a, (y - a) + dy]$ by $dy - a$:

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\{X + a \in dy\} = \mathbf{P}\{X \in dy - a\}.$$

Finally, we calculate this quantity in terms of the density f . Since $dy - a$ is an interval around $y - a$, we look at the value $f(y - a)$. On the other hand, since the interval $dy - a$ has length dy , we multiply $f(y - a)$ by dy :

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\{X \in dy - a\} = f(y - a)dy.$$

This is a very intuitive calculation if you think about the interpretation of the dx notation as discussed after (5).

3.1.2 Scaling by a positive number

Suppose that X is a continuous random variable with density f and define $Y = cX$ for some $c > 0$. To compute the density of Y , that is,

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\{cX \in dy\},$$

we argue as in the case of shifting:

$$\begin{aligned} \mathbf{P}\{Y \in dy\} &= \mathbf{P}\{cX \in [y, y + dy]\} \\ &= \mathbf{P}\{y \leq cX \leq y + dy\} \\ &= \mathbf{P}\left\{\frac{y}{c} \leq X \leq \frac{y}{c} + \frac{dy}{c}\right\} \\ &= \mathbf{P}\left\{X \in \left[\frac{y}{c}, \frac{y}{c} + \frac{dy}{c}\right]\right\}. \end{aligned}$$

While the length of the interval does not change by shifting, it is shrunk by a factor of c here. We will denote the interval $[\frac{y}{c}, \frac{y}{c} + \frac{dy}{c}]$ by $\frac{dy}{c}$:

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\{cX \in dy\} = \mathbf{P}\left\{X \in \frac{dy}{c}\right\}.$$

Let us compute this quantity in terms of the density f . Since $\frac{dy}{c}$ is an interval around $\frac{y}{c}$, we look at the value $f(\frac{y}{c})$. Since the length of the interval $\frac{dy}{c}$ is $\frac{dy}{c}$, we multiply $f(\frac{dy}{c})$ by $\frac{dy}{c}$:

$$\mathbf{P}\{Y \in dy\} = \mathbf{P}\left\{X \in \frac{dy}{c}\right\} = f\left(\frac{y}{c}\right) \frac{dy}{c}.$$

Remark 3 (Naïve and wrong). Note that one might naively attempt writing one of

$$\begin{aligned}\mathbf{P}\{Y \in dy\} &= \mathbf{P}\{cX \in dy\} = \mathbf{P}\left\{X \in \frac{dy}{c}\right\} \stackrel{?}{=} f(y) \frac{dy}{c}, \\ \mathbf{P}\{Y \in dy\} &= \mathbf{P}\{cX \in dy\} \stackrel{?}{=} \mathbf{P}\left\{X \in d\left(\frac{y}{c}\right)\right\} \stackrel{?}{=} f\left(\frac{y}{c}\right) dy,\end{aligned}$$

both of which are wrong!

Remark 4 (What is wrong with negative coefficients?). You might wonder why the above method for positive coefficient does not work for negative ones. When we try to find the distribution of $-cX$ ($c > 0$) by mimicing the above method, note that

$$\begin{aligned}\mathbf{P}\{-cX \in [y, y + dy]\} &= \mathbf{P}\{y \leq -cX \leq y + dy\} \\ &= \mathbf{P}\left\{-\frac{y}{c} - \frac{dy}{c} \leq X \leq -\frac{y}{c}\right\} \\ &= \mathbf{P}\left\{X \in \left[-\frac{y}{c} - \frac{dy}{c}, -\frac{y}{c}\right]\right\}.\end{aligned}$$

In the second equality, the ordering of the quantities is switched due to multiplication by a negative constant. The resulting interval, $[-\frac{y}{c} - \frac{dy}{c}, -\frac{y}{c}]$ has a different form than our usual form $dx = [x, x + dx]$, hence we avoid manipulations with negative constants!

3.2 A method: change of variables

In general, we might want to compute the distributions of functions of random variables that are not limited to shifting and scaling with positive numbers. We will describe a general method that works for one-to-one functions of random variables (and some more general functions that can be reduced to the one-to-one case).

Before we describe this useful method, let us phrase a result which will be the basis of the method.

Proposition 5. Suppose that Y is a real-valued random variable and $g: \mathbb{R} \rightarrow \mathbb{R}_+$ is a function with $\int_{-\infty}^{\infty} g(x)dx = 1$. Then, g is the density of Y if and only if we have

$$\mathbf{E}(\varphi(Y)) = \int_{-\infty}^{\infty} \varphi(y)g(y)dy, \quad (7)$$

for every (“nice”, e.g. bounded, continuous) function φ .

There is nothing new about this result. Indeed, if g is the density of Y , then the expectation $\mathbf{E}(\varphi(Y))$ is already defined by (7). On the other hand, if (7) holds for every (bounded) function φ , then we can take $\varphi(y) = \mathbf{1}_{[a,b]}(y)$ and get

$$\mathbf{P}\{Y \in [a, b]\} = \mathbf{E}(\mathbf{1}_{[a,b]}(Y)) = \int_{-\infty}^{\infty} \mathbf{1}_{[a,b]}(y)f(y)dy = \int_a^b g(y)dy,$$

for every $a, b \in \mathbb{R}$ with $a \leq b$. But this means that g is the density of Y .

Next, we describe the method of computing the distribution of $Y = h(X)$, where X is a continuous random variable with (known) density f . Let us try to compute $\mathbf{E}(\varphi(Y))$ for an arbitrary function φ . Note that

$$\varphi(Y) = \varphi(h(X)),$$

which is a function of X . Hence,

$$\mathbf{E}(\varphi(Y)) = \mathbf{E}(\varphi(h(X))) = \int_{-\infty}^{\infty} \varphi(h(x))f(x)dx.$$

The whole point of the method is to make a sequence of substitutions in the integral to convert it to the form

$$\mathbf{E}(\varphi(Y)) = \int_{-\infty}^{\infty} \varphi(h(x))f(x)dx = \dots (\text{substitutions}) \dots = \int_{-\infty}^{\infty} \varphi(y)g(y)dy$$

for some function $g: \mathbb{R} \rightarrow \mathbb{R}_+$ with $\int_{-\infty}^{\infty} g(y)dy = 1$ (that is free of the choice of φ). Since φ is an arbitrarily chosen, we have

$$\mathbf{E}(\varphi(Y)) = \int_{-\infty}^{\infty} \varphi(y)g(y)dy$$

for every function φ . By Proposition 5, we conclude that g is the density of Y .

Example 6. Let us verify our result for scaling with positive numbers using this method. If $Y = cX$ for some $c > 0$, then

$$\begin{aligned} \mathbf{E}(\varphi(Y)) &= \mathbf{E}(\varphi(cX)) \\ &= \int_{-\infty}^{\infty} \varphi(cx)f(x)dx \\ &= \int_{-\infty}^{\infty} \varphi(y)f\left(\frac{y}{c}\right)\frac{dy}{c}, \end{aligned}$$

where we make the substitution $y = 2x$ to get the third equality. Hence, we conclude that

$$\mathbf{P}\{Y \in dy\} = f\left(\frac{y}{2}\right) \frac{dy}{2}.$$

Example 7. Let X be a uniformly distributed random variable in the interval $[a, b]$, where $0 < a < b$. Let us find the density of $Y = X^2$. For an arbitrary function φ , we have

$$\begin{aligned} \mathbf{E}(\varphi(Y)) &= \mathbf{E}(\varphi(X^2)) \\ &= \int_a^b \varphi(x^2) \frac{1}{b-a} dx \\ &= \int_{a^2}^{b^2} \varphi(y) \frac{1}{2(b-a)\sqrt{y}} dy \\ &= \int_{-\infty}^{\infty} \varphi(y) \mathbf{1}_{[a^2, b^2]}(y) \frac{1}{2(b-a)\sqrt{y}} dy, \end{aligned}$$

where we make the substitution $y = x^2$ (hence $dy = 2x dx$ and $dx = \frac{dy}{2\sqrt{y}}$) to get the last equality. Hence, the density of Y is given by

$$\mathbf{P}\{Y \in dy\} = \frac{1}{2(b-a)\sqrt{y}} dy, \quad y \in [a^2, b^2],$$

or equivalently,

$$\mathbf{P}\{Y \in dy\} = \mathbf{1}_{[a^2, b^2]}(y) \frac{1}{2(b-a)\sqrt{y}} dy, \quad y \in \mathbb{R}.$$

(Pay attention to the extremes of integration. They are part of the definition of the density function!)

Example 8. Let $X \sim N(0, 1)$ and $Y = X^2$. For an arbitrary function φ , we have

$$\mathbf{E}(\varphi(Y)) = \mathbf{E}(\varphi(X^2)) = \int_{-\infty}^{\infty} \varphi(x^2) \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx.$$

Note that $x \mapsto \varphi(x) = x^2$ is not one-to-one on the real line, so we split the integral into two parts where φ is one-to-one and hence the substitution works:

$$\begin{aligned} \mathbf{E}(\varphi(Y)) &= \mathbf{E}(\varphi(X^2)) \\ &= \int_{-\infty}^0 \varphi(x^2) \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx + \int_0^{\infty} \varphi(x^2) \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx \\ &= 2 \int_0^{\infty} \varphi(x^2) \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx \\ &= 2 \int_0^{\infty} \varphi(y) \frac{e^{-\frac{y}{2}}}{\sqrt{2\pi}} \frac{dy}{2\sqrt{y}} \\ &= \int_0^{\infty} \varphi(y) \frac{e^{-\frac{y}{2}}}{\sqrt{2\pi y}} dy. \end{aligned}$$

Hence, the density of Y is given by

$$\mathbf{P}\{Y \in dy\} = \frac{e^{-\frac{y}{2}}}{\sqrt{2\pi y}} dy, \quad y \geq 0.$$

This distribution is sometimes called the *chi-square distribution*.

3.3 Change of variables for random vectors

The above method to compute densities works for random vectors as well. Let us first state the two-dimensional version of Proposition 5.

Proposition 9. *Suppose that (U, V) is a random vector and $g: \mathbb{R}^2 \rightarrow \mathbb{R}_+$ is a function with $\int_{u,v \in \mathbb{R}} g(u, v) du dv = 1$. Then, g is the joint density of U, V if and only if we have*

$$\mathbf{E}(\varphi(U, V)) = \int_{u,v \in \mathbb{R}} \varphi(u, v) g(u, v) du dv,$$

for every (“nice”, e.g. bounded, continuous) function $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}$.

Suppose that we are given a random vector (X, Y) with density f and let $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function. Define the random vector $(U, V) = h(X, Y)$. Using the multidimensional version of the change of variables method, we will compute the joint density of U, V . To that end, let $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}$ be an arbitrary function. We have

$$\mathbf{E}(\varphi(U, V)) = \mathbf{E}(\varphi(h(X, Y))) = \int_{x,y \in \mathbb{R}} \varphi(h(x, y)) f(x, y) dx dy.$$

The method consists of making changes of variables to get

$$\begin{aligned} \mathbf{E}(\varphi(U, V)) &= \int_{x,y \in \mathbb{R}} \varphi(h(x, y)) f(x, y) dx dy \\ &= \dots (\text{substitutions}) \dots \\ &= \int_{u,v \in \mathbb{R}} \varphi(u, v) g(u, v) du dv \end{aligned}$$

for some function $g: \mathbb{R}^2 \rightarrow \mathbb{R}_+$ with $\int_{u,v \in \mathbb{R}} g(u, v) du dv = 1$. Once this is done, we conclude that g is the joint density of U, V thanks to Proposition 9.

Example 10 (Gaussian coordinates polarized). Suppose that we pick a random point on the plane with (Cartesian) coordinates (X, Y) and assume that X, Y are independent standard Gaussian random variables. The polar coordinates of this random point are given by the pair $(\sqrt{U}, V)$, where $U = X^2 + Y^2$ and $V = \alpha(X, Y)$ is the positive angle between the x -axis and the line passing through (X, Y) and the origin. Note that $\tan(V) = \frac{Y}{X}$. We will determine the joint distribution of U and V .

Let us take an arbitrary function $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}$. We have

$$\begin{aligned}\mathbf{E}(\varphi(U, V)) &= \mathbf{E}(\varphi(X^2 + Y^2, \alpha(X, Y))) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi(x^2 + y^2, \alpha(x, y)) \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}} dy dx \\ &= \left(\int_{-\infty}^0 + \int_0^{\infty} \right) \int_{-\infty}^{\infty} \varphi(x^2 + y^2, \alpha(x, y)) \frac{e^{-\frac{x^2+y^2}{2}}}{2\pi} dy dx.\end{aligned}$$

At this point, we make the substitution $v = \alpha(x, y)$ in the inner integral (with respect to dy) for each fixed value of x . If $x > 0$, then v runs through the set $(\frac{3\pi}{2}, 2\pi] \cup (0, \frac{\pi}{2})$ to as y runs through the set $(-\infty, \infty)$. If $x < 0$, then v runs from $\frac{3\pi}{2}$ back to $\frac{\pi}{2}$ as y runs through the set $(-\infty, \infty)$. Note that the substitution results in $y = x \tan(v)$ and $dy = x(1 + \tan^2(v))dv$. Therefore,

$$\begin{aligned}\mathbf{E}(\varphi(U, V)) &= \int_0^{\infty} \left(\int_{\frac{3\pi}{2}}^{2\pi} + \int_0^{\frac{\pi}{2}} \right) \varphi(x^2(1 + \tan^2(v)), v) \frac{e^{-\frac{1}{2}x^2(1 + \tan^2(v))}}{2\pi} x(1 + \tan^2(v)) dv dx \\ &\quad + \int_{-\infty}^0 \int_{\frac{3\pi}{2}}^{\frac{\pi}{2}} \varphi(x^2(1 + \tan^2(v)), v) \frac{e^{-\frac{1}{2}x^2(1 + \tan^2(v))}}{2\pi} x(1 + \tan^2(v)) dv dx \\ &= \int_0^{\infty} \left(\int_{\frac{3\pi}{2}}^{2\pi} + \int_0^{\frac{\pi}{2}} \right) \varphi(x^2(1 + \tan^2(v)), v) \frac{e^{-\frac{1}{2}x^2(1 + \tan^2(v))}}{2\pi} x(1 + \tan^2(v)) dv dx \\ &\quad + \int_0^{\infty} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \varphi(x^2(1 + \tan^2(v)), v) \frac{e^{-\frac{1}{2}x^2(1 + \tan^2(v))}}{2\pi} x(1 + \tan^2(v)) dv dx \\ &= \int_0^{\infty} \int_0^{2\pi} \varphi(x^2(1 + \tan^2(v)), v) \frac{e^{-\frac{1}{2}x^2(1 + \tan^2(v))}}{2\pi} x(1 + \tan^2(v)) dv dx \\ &= \int_0^{2\pi} \left(\int_0^{\infty} \varphi(x^2(1 + \tan^2(v)), v) e^{-\frac{1}{2}x^2(1 + \tan^2(v))} x dx \right) \frac{1 + \tan^2(v)}{2\pi} dv.\end{aligned}$$

Next, we make the substitution $u = x^2(1 + \tan^2(v))$ in the inner integrals for every fixed v . As x runs through the set $[0, \infty)$, u runs through the set $[0, \infty)$. The substitution results in $x dx = \frac{du}{2(1 + \tan^2(v))}$. Hence,

$$\begin{aligned}\mathbf{E}(\varphi(U, V)) &= \int_0^{2\pi} \left(\int_0^{\infty} \varphi(u, v) e^{-\frac{u}{2}} \frac{1}{2(1 + \tan^2(v))} du \right) \frac{1 + \tan^2(v)}{2\pi} dv \\ &= \int_0^{2\pi} \int_0^{\infty} \varphi(u, v) \cdot \left(\frac{1}{2} e^{-\frac{u}{2}} du \right) \cdot \left(\frac{1}{2\pi} dv \right).\end{aligned}$$

Therefore,

$$\mathbf{P}\{U \in du, V \in dv\} = \frac{1}{2} e^{-\frac{u}{2}} du \cdot \frac{1}{2\pi} dv, \quad u \geq 0, v \in (0, 2\pi).$$

Does anything look familiar in this joint density? Note that $v \mapsto \frac{1}{2\pi}$ is the density of $\text{Unif}(0, 2\pi)$ and $u \mapsto \frac{1}{2}e^{-\frac{u}{2}}$ is the density of $\text{Expon}(\frac{1}{2})$. Can we conclude that $V \sim \text{Unif}(0, 2\pi)$ and $U \sim \text{Expon}(\frac{1}{2})$? If so, can we also conclude that U, V are independent?

The answer is “yes” to all questions! Note that we have

$$\mathbf{E}(\varphi(U, V)) = \int_0^{2\pi} \int_0^\infty \varphi(u, v) \cdot \left(\frac{1}{2}e^{-\frac{u}{2}} du\right) \cdot \left(\frac{1}{2\pi} dv\right) \quad (8)$$

for every function $\varphi(u, v)$. In particular, this is true for every function φ that does not depend on v : $\varphi(u, v) = \phi(u)$. In this case, we get

$$\begin{aligned} \mathbf{E}(\phi(U)) &= \mathbf{E}(\varphi(U, V)) \\ &= \int_0^{2\pi} \left(\int_0^\infty \phi(u) \cdot \frac{1}{2}e^{-\frac{u}{2}} du \right) \frac{1}{2\pi} dv \\ &= \left(\int_0^\infty \phi(u) \frac{1}{2}e^{-\frac{u}{2}} du \right) \left(\int_0^{2\pi} \frac{1}{2\pi} dv \right) \\ &= \int_0^\infty \phi(u) \frac{1}{2}e^{-\frac{u}{2}} du \end{aligned}$$

for every function ϕ . By Proposition 5, we conclude that $U \sim \text{Expon}(\frac{1}{2})$. Similarly, we can check that $V \sim \text{Unif}(0, 2\pi)$. Because the joint density in (8) is given as the product of the marginal densities of U and V , we conclude that U and V are independent.

4 Conditioning

We finish these notes by reminding the use of the dx notation when computing conditional probabilities and expectations.

Let A, B be events. If $\mathbf{P}(B) > 0$, then we define

$$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} \quad (9)$$

Here $A \cap B$ means both A and B occur. Sometimes we replace $\cap$ with a comma to indicate the same meaning such as $\mathbf{P}\{X = 1, Y = 2\}$.

4.1 Conditioning: discrete random variables

Consider the case $A = \{X = a_i\}$, where X is a discrete random variable. By definition,

$$\mathbf{P}(\{X = a_i\}|B) = \frac{\mathbf{P}(\{X = a_i\} \cap B)}{\mathbf{P}(B)}.$$

In particular, assume that we have another discrete random variable Y with values in $\{b_1, b_2, \dots\}$, and suppose that $B = \{Y = b_j\}$. Then

$$\mathbf{P}(\{X = a_i\}|B) = \mathbf{P}\{X = a_i|Y = b_j\} = \frac{\mathbf{P}\{X = a_i, Y = b_j\}}{\mathbf{P}\{Y = b_j\}}.$$

Once we know the joint distribution of X, Y , these quantities can be computed easily.

4.2 Conditioning: continuous random variables

The dx notation makes it possible to formulate easily the case where X is a continuous random variable. In this case, the *conditional density* of X given B is defined by

$$\mathbf{P}(\{X \in dx\}|B) = \frac{\mathbf{P}(\{X \in dx\} \cap B)}{\mathbf{P}(B)}.$$

Recall that $dx = [x, x + dx]$ should be thought of as an infinitesimal interval around x with length dx .

If we have a continuous random variable Y and we let $B = \{Y = y\}$, then we have $\mathbf{P}\{Y = y\} = 0$ leading to

$$\mathbf{P}\{X \in dx|Y = y\} = \frac{\mathbf{P}\{X \in dx, Y = y\}}{\mathbf{P}\{Y = y\}} = \frac{0}{0}.$$

This is not well-defined! The correct definition of $\mathbf{P}\{X \in dx|Y = y\}$ has the following “trick” using the dx notation:

$$\mathbf{P}\{X \in dx|Y = y\} := \frac{\mathbf{P}\{X \in dx, Y \in dy\}}{\mathbf{P}\{Y \in dy\}}.$$

According to this definition, if we condition on $\{Y = y\}$ where Y is a continuous random variable, we can picture $|Y = y$ the same as $|Y \in dy$ in our mind and apply the “usual” rules of computing conditional probabilities. However, in the expressions we still have to use the version $|Y = y$: the right hand side of the above computation has one dy in the numerator and one dy in the denominator, hence they cancel out as a result of our computations and the final answer is free of any dy s! So there is no reason to have a dy on the left hand side!

When X, Y are continuous random variables, the law of total probability becomes

$$\mathbf{P}\{X \in dx\} = \int_y \mathbf{P}\{X \in dx|Y = y\}\mathbf{P}\{Y \in dy\},$$

and Bayes’ rule becomes

$$\mathbf{P}\{Y \in dy|X = x\} = \frac{\mathbf{P}\{X \in dx|Y = y\}\mathbf{P}\{Y \in dy\}}{\int_y \mathbf{P}\{X \in dx|Y = y\}\mathbf{P}\{Y \in dy\}}.$$

Note that the previous derivations with the dx notation are heuristic. If you are interested to know the rigorous mathematical foundations behind them, you are welcome to take a measure-theoretic probability course such as MAT 385 and ORF 526.

4.3 Conditioning: mixed types

If X is discrete and Y is continuous, then we have

$$\mathbf{P}\{X = a_i | Y = y\} = \frac{\mathbf{P}\{X = a_i, Y \in dy\}}{\mathbf{P}\{Y \in dy\}}$$

and Bayes' formula, law of total probability, etc. can be written similarly.

4.4 Conditional expectation

When we look at the definition of conditional probability (9), we can heuristically think of

$$\mathbf{P}_B(A) := \mathbf{P}(A|B) \quad \text{or} \quad \mathbf{P}_B(\cdot) = \mathbf{P}(\cdot|B).$$

This is roughly seeing $\mathbf{P}(\cdot|B)$ as a new probability measure. By conditioning, we have extra information of certain events and that changes our perception of the probabilities.

Following this intuition, the conditional expectation of a random variable X given B is defined as

$$\begin{aligned} \mathbf{E}(X|B) &= \sum_{i=1}^{\infty} a_i \mathbf{P}(\{X = a_i\}|B), \\ \mathbf{E}(X|B) &= \int_{-\infty}^{\infty} x \mathbf{P}(\{X \in dx\}|B), \end{aligned}$$

when X is discrete or continuous, respectively. You can compare this definition with the ones in section 1, and see how the $|B$ kicks in.

Furthermore, for a random variable $X \geq 0$, the formula

$$\mathbf{E}(X) = \int_0^{\infty} \mathbf{P}\{X > t\} dt$$

has a conditional version

$$\mathbf{E}(X|B) = \int_0^{\infty} \mathbf{P}(\{X > t\}|B) dt,$$

as we have already seen in Reliability Theory. Here, this B can be anything we have seen: $\{Y = 1\}$ when Y is discrete or continuous, or events like $\{1 < Y < 2\}$.